K.R. Mangalam University

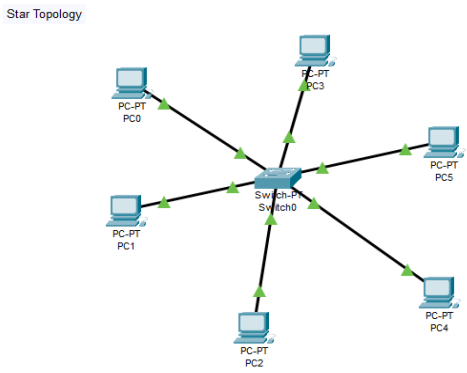
Computer Networks Lab Assignment

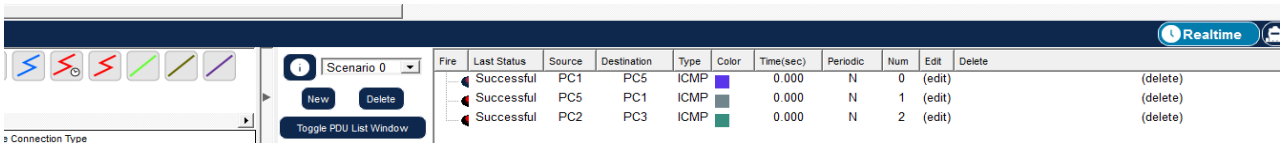
Submitted By:
AKSHAT SHARMA
2301420027
B.Tech CSE (DS)

Faculty:
Dr. Shahid Ahmad Wani
Computer Networks
(SOET)

Topologies

1. Star Topology





Brief = All devices are connected to a central device (switch/hub). Communication always passes through the central node.

Setup

- **1 Switch**
- **5-6 PCs**
- **Straight-through cables**

Working (In Depth)

- **When PC1 sends data to PC5 → data goes to switch**
- **Switch checks MAC address table**
- **Forwards data only to destination port**
- **Reduces collisions (compared to hub)**

Advantages

- **Easy to install & manage**
- **Fault isolation is simple**
- **Adding/removing nodes is easy**

Disadvantages

- **Switch failure = entire network down**
- **More cable required**

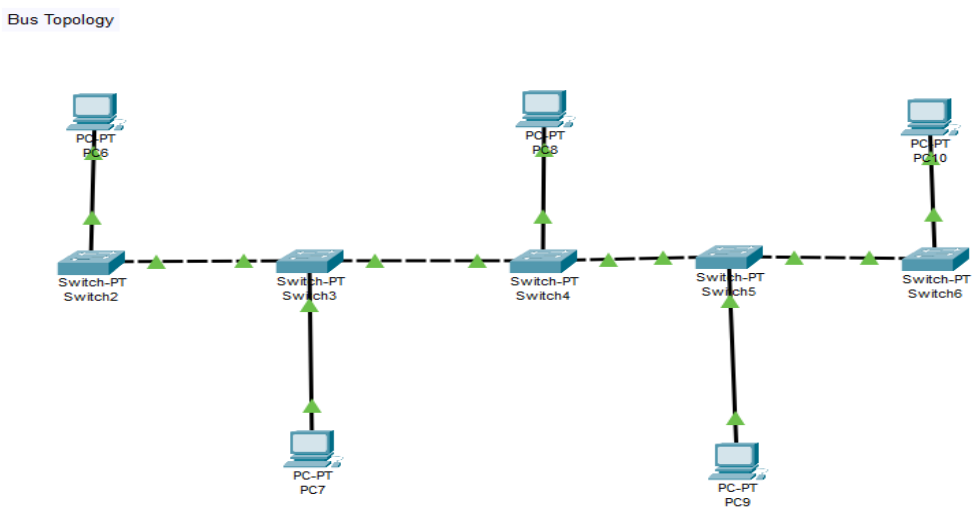
Observations:

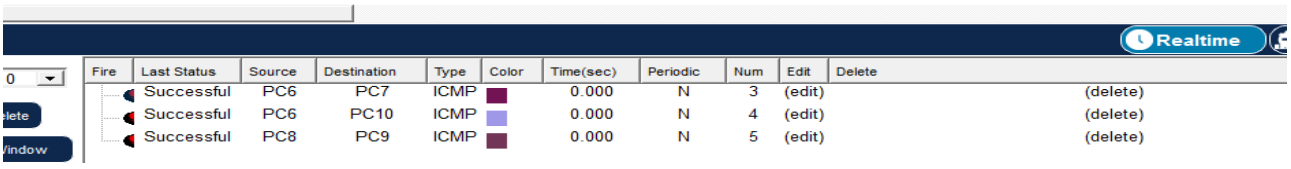
- All PCs are connected to a central switch, and **no direct communication exists between end devices**.
- During simulation, when a ping (ICMP) is sent, the packet first reaches the switch and then is forwarded to the destination device.
- The switch uses its **MAC address table** to send data only to the correct port, reducing unnecessary traffic.
- Data transmission is **fast and efficient**, with minimal collisions.
- If one PC is disconnected, the rest of the network continues to function normally.
- However, when the switch is turned off or fails, **all communication in the network stops completely**.
- The topology shows **centralized control**, making monitoring and troubleshooting easier.using **ping command** (all devices responded successfully)

Result

Star topology provides efficient, centralized communication with easy management.

2. Bus Topology





Realtime

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC6	PC7	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC6	PC10	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC8	PC9	ICMP		0.000	N	5	(edit)	(delete)

Brief = All devices are connected to a single backbone cable (linear structure).

Setup

- **Multiple switches in line**
- **PCs connected to switches**

Working (In Depth)

- **Data is sent along the backbone**
- **Every device receives the signal**
- **Only the intended device accepts it**
- **Others ignore it**

Advantages

- **Low cost**
- **Simple design**
- **Uses less cable**

Disadvantages

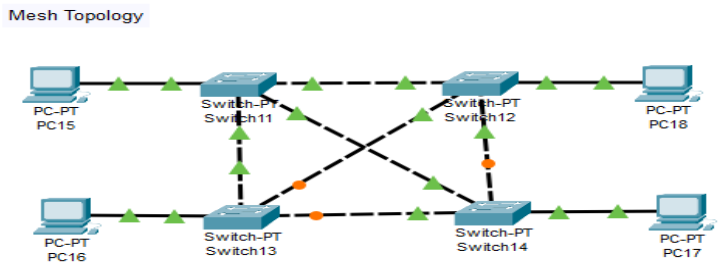
- **Backbone failure → entire network fails**
- **More collisions**
- **Difficult troubleshooting**

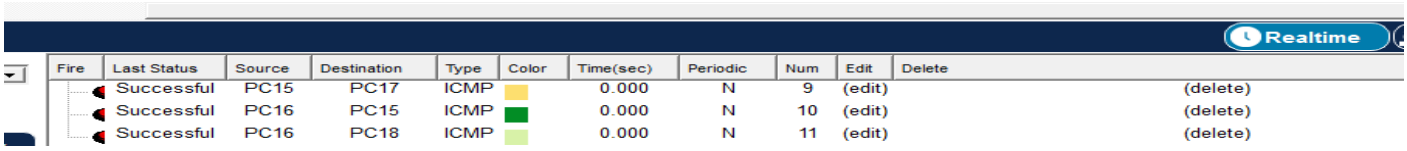
Observations:

- All devices are connected along a single backbone line, and **data travels in a linear path**.
- When a device sends data, the signal is **broadcast to all devices**, but only the destination device accepts it.
- Other devices receive the packet but ignore it after checking the address.
- During simulation, it is observed that **network traffic increases** because all devices see the data.
- Communication is successful as long as the backbone is intact.
- If any point in the backbone cable is disconnected, **the entire network communication fails**.
- The topology shows **higher chances of collision** due to shared medium.

Result = Bus topology is cost-effective but less reliable due to single point failure.

3. **Mesh Topology**





Realtime

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC15	PC17	ICMP		0.000	N	9	(edit)	(delete)
	Successful	PC16	PC15	ICMP		0.000	N	10	(edit)	(delete)
	Successful	PC16	PC18	ICMP		0.000	N	11	(edit)	(delete)

Brief - Every device is connected to every other device (multiple paths).

Setup

- Multiple switches
- Fully interconnected
- PCs connected

Working (In Depth)

- Data can take multiple paths
- If one path fails → alternate path used
- Routing ensures best path selection

Advantages

- Highly reliable
- Fault tolerant
- No single point of failure

Disadvantages

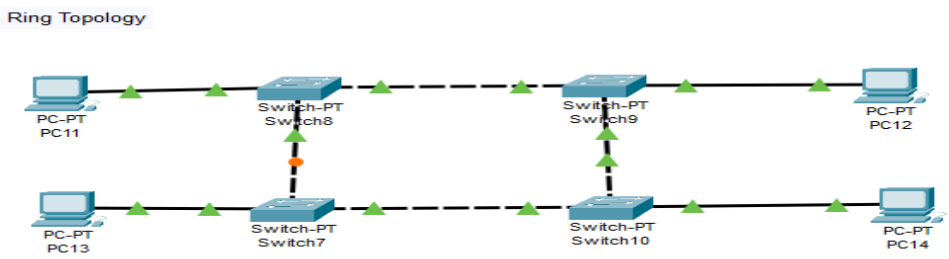
- Expensive
- Complex wiring
- Hard to maintain

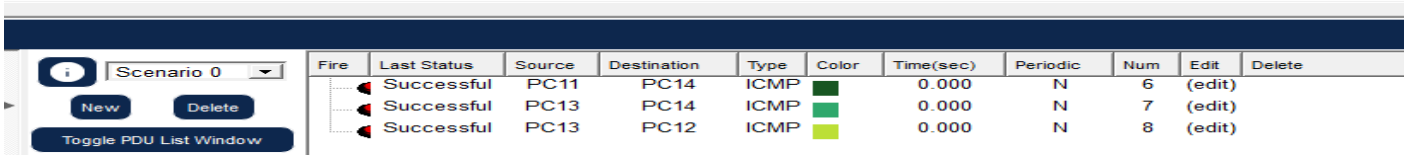
Observations:

- Each device is connected to multiple other devices, forming **multiple communication paths**.
- During simulation, data can travel through **different routes** to reach the destination.
- If one link fails, packets automatically use an **alternate path**, ensuring uninterrupted communication.
- The network shows **high reliability and fault tolerance**.
- Traffic is distributed across multiple links, reducing congestion.
- However, a large number of cables and connections make the network **complex and expensive**.
- The topology demonstrates **no single point of failure**, making it suitable for critical systems.

Result = Mesh topology provides maximum reliability and redundancy**, ideal for critical systems.**

4. **Ring Topology**





Brief = Devices are connected in a **circular loop**, each device connected to two others.

Setup

- Switches connected in circle
- PCs attached

Working (In Depth)

- Data travels in one direction (clockwise/anticlockwise)
- Uses **token passing mechanism**
- Only device with token can send data
- Reduces collisions

Advantages

- No collisions
- Equal access for all nodes

Disadvantages

- One link failure → entire network stops
- Difficult to reconfigure

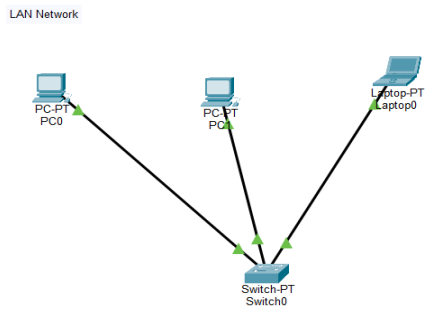
Result = Ring topology ensures **organized data flow**, but is sensitive to failures.

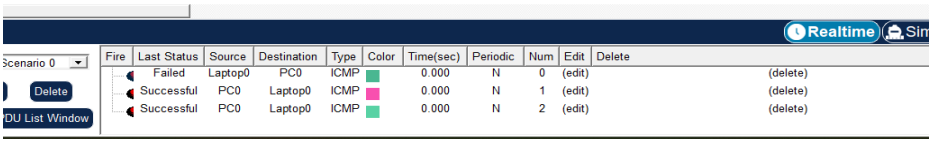
Observations:

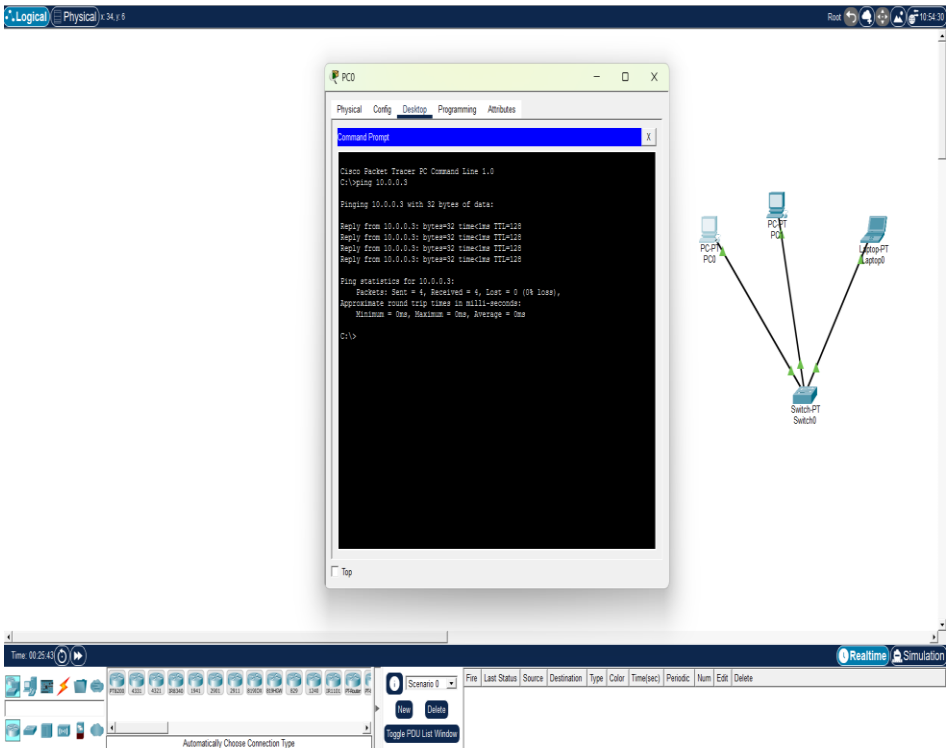
- Devices are connected in a circular loop, and **each device is connected to exactly two neighbors**.
- Data travels in a **fixed direction (clockwise or anticlockwise)** across the ring.
- During transmission, data passes through each intermediate device until it reaches the destination.
- The concept of **token passing** ensures that only one device transmits at a time, reducing collisions.
- In simulation, packets are seen moving node-to-node in a circular manner.
- If any one link or device fails, **the entire ring breaks and communication stops**.
- The topology ensures **equal access to all devices**, but lacks flexibility.

Types of Networks

1. LAN Network







Brief = A LAN connects devices within a **small geographical area** such as a room, lab, or building.

Working (In Depth)

- All PCs are connected to a **single switch (Switch0)**
- Communication happens using **MAC addresses (Layer 2)**
- When PC0 sends data to PC1:
  - Packet goes to switch
  - Switch forwards to correct port
- No router is required since all devices are in same network

Observation

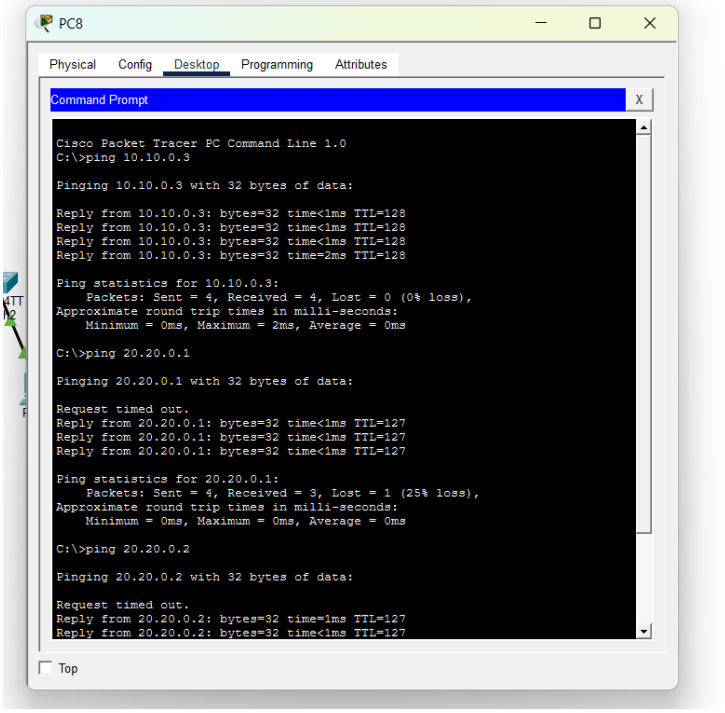
- All devices (PC0, PC1, Laptop0) are connected to a **central switch forming a star topology**
- When ping command (ping 10.0.0.3) is executed:
  - **All packets are successfully received (0% loss)**
  - Response time is **<1ms**, showing fast communication
- The switch forwards frames only to destination port → **efficient communication**
- No network delay or congestion observed
- Communication is limited within the same network
- No routing is involved (pure switching environment)

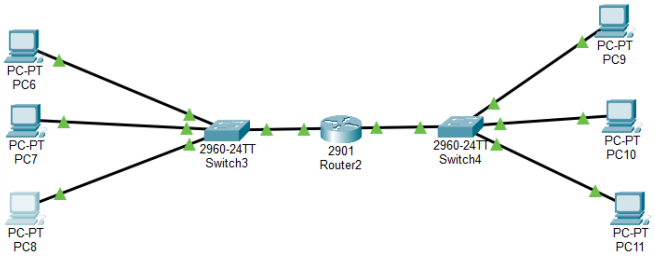
Result

LAN was successfully implemented.

All devices communicated efficiently with **high speed, low latency, and zero packet loss**.

2. MAN Network





Brief = A MAN connects multiple LANs within a **city or large campus**.

Working (In Depth)

- Two LANs are connected using a **router (Router2)**
- Each LAN has its own IP range:
  - LAN1 → 10.10.0.x
  - LAN2 → 20.20.0.x
- Router performs:
  - **IP routing (Layer 3)**
  - Decides best path for packet transmission

Observation

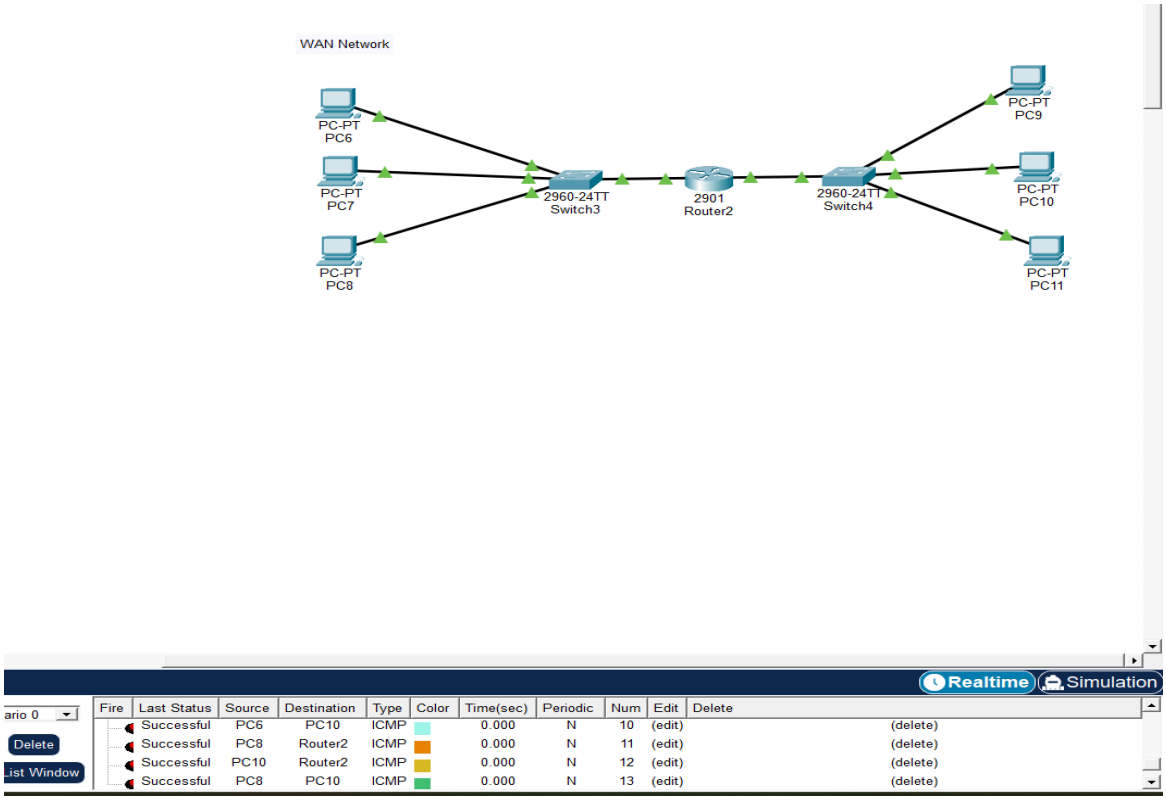
- Devices are divided into **two separate networks connected via router**
- When pinging within same LAN:
  - Communication is **instant and successful**
- When pinging across networks:

    - First request shows **timeout (initial delay)**
    - After that, replies are successful
    - This delay occurs due to **ARP resolution**
- Router forwards packets between networks → **inter-network communication achieved**
- TTL value reduces (128 → 127), indicating packet passed through router
- Slight delay compared to LAN due to routing process

Result

MAN was successfully implemented.
Different networks communicated through a router with **slight delay but successful transmission.**

3. **WAN Network**



Brief = A WAN connects networks over a **large geographical area** (country or world).

Working (In Depth)

- Multiple LANs are connected through **routers and switches**
- Communication happens via:

    - Routing protocols
    - IP addressing
- Data travels across multiple networks before reaching destination

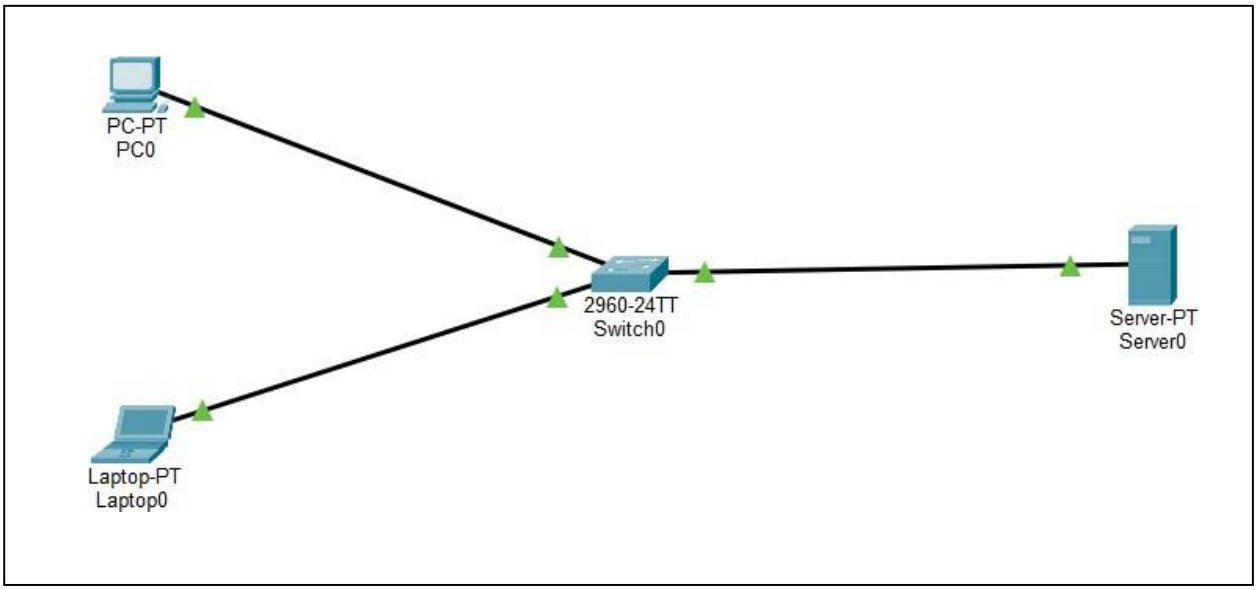
Observation

- Two distant LANs are connected via a router forming a WAN
- Communication between PCs on different sides is **successful**
- Ping results show:
    - Slight delay compared to LAN and MAN
    - Initial packet loss due to ARP / routing setup
- Router plays a crucial role in forwarding packets between networks
- Network shows **scalability**, as more LANs can be added
- More complexity observed compared to LAN
- Data travels longer path → **increased latency**

Result

WAN was successfully implemented.
Communication across distant networks was achieved with **higher latency but reliable data transmission**.

DHCP + Email configuration



A DHCP server was configured to automatically assign IP addresses to devices in the network. An email server was also set up to enable communication between users. The experiment demonstrated automated network configuration and basic client-server communication.

Observations:

- Used 1 server, 1 switch, 1 PC, and 1 laptop
- Configured server as DHCP server for automatic IP assignment
- Enabled SMTP and POP3 services for email communication
- Verified dynamic IP allocation and successful email exchange